LESSON 8.7

Equations and Inequalities Problem Solving

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can model and solve real-world problems using equations and inequalities.

WHY THIS MATTERS Detective Santos is closing a complex case. She has collected equations and inequalities from different parts of the investigation. One clue says: 'The total value of stolen items divided equally among 4 accomplices gave each person $85.' Another says: 'The lookout earned less than $50.' She must model both clues mathematically and check if the answers are reasonable. Can you help?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Model *Spanish: Modelar*	To show a real-life situation with an equation or inequality.	'Twice a number is 18' → 2n = 18 — the equation models the words
Equation *Spanish: Ecuación*	A math sentence with an equal sign showing both sides are the same.	x + 8 = 15 — exactly one answer: x = 7
Inequality *Spanish: Desigualdad*	A math sentence that compares two sides with <, >, ≤, or ≥.	x + 5 > 12 — many answers: x = 8, 9, 10, ...
Reasonableness *Spanish: Razonabilidad*	Checking if your answer makes sense.	If the problem asks for a number of people and you get x = −3, that is NOT reasonable
Variable *Spanish: Variable*	A letter that stands for a number that is unknown or can change.	“5 less than a number is 12” becomes x − 5 = 12, where x is the variable.
Constraint *Spanish: Restricción*	A limit in a problem that tells what values are allowed.	“You can spend at most $20” is a constraint: cost ≤ 20.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Clue: 'A total divided equally among 4 people gave each $85.'

1. This has an exact total, so I use an equation. Let t be the total.
2. Divided among 4 to get 85 means t ÷ 4 = 85, so t = 85 × 4 = 340.
3. Check reasonableness: $340 split 4 ways is $85 each. That makes sense!

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Clue: 'The lookout earned less than $50.' Equation or inequality?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I use an equation when the problem says _____, but I use an inequality when the problem says _____ because _____.

Word bank Model | Equation | Inequality | Reasonableness | Variable | Constraint | equation | inequality | equal | at least | at most | more than

WATCH OUT!

A common mistake

A common mistake in Equations and Inequalities Problem Solving is matching the wrong model to a key phrase. For 'the getaway car held no more than 6 bags,' students often write the equation $b = 6$ instead of the inequality $b \leq 6$, because they see one number and assume it must be exact. The same mistake happens in reverse: for 'the reward split equally among 5 people gave each $40,' students sometimes write the inequality $5p \geq 40$ instead of the equation $5p = 40$, because 'split equally' sounds like it could mean a range. Before you submit, ask: does this phrase give one exact value (equation, =) or a limit/range (inequality, <, >, ≤, ≥)?

LESSON 8.7

Equations and Inequalities Problem Solving

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which model is correct for: 'A number divided by 6 equals 7'?

(A) n / 6 = 7
(B) 6n = 7
(C) n − 6 = 7
(D) n + 6 = 7

HINT Spot the operation phrase: 'divided by 6'. That phrase gives you the symbol between n and 6.

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Sort each problem: equation or inequality?

Groups: Equation (=) | Inequality (<, >, ≤, ≥)

_____ Five boxes weigh exactly 60 pounds total. How much does each weigh?

_____ A number plus 8 equals 20.

_____ A shelf holds at most 30 books. There are already 17. How many more fit?

_____ The temperature must stay above 32 degrees.

HINT 'Exactly,' 'equals,' and 'is' signal an equation.

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A museum has some paintings. After adding 12, they have 45. Which equation models this?

(A) $p + 12 = 45$

(B) $p - 12 = 45$

(C) $12p = 45$

(D) $p / 12 = 45$

HINT The museum STARTS with an unknown number of paintings p, then 12 more are ADDED to reach 45.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A detective needs more than 8 hours to finish the investigation. She has already worked 3 hours. Which inequality represents the additional hours h she needs?

(A) $3 + h > 8$

(B) $3 + h = 8$

(C) $h > 3$

(D) $8 + h > 3$

HINT There are two time pieces: the 3 hours already worked and the extra hours h. Together they must pass 8 hours.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A bus can carry at most 48 passengers. There are already 31 on board. Which inequality shows how many more can board?

(A) $31 + p \leq 48$

(B) $31 + p > 48$

(C) $31 + p = 48$

(D) $p - 31 \leq 48$

HINT The bus already has 31 riders, and p more may board — but the TOTAL cannot pass 48.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Problem** A detective already found 8 fingerprints at the scene. After dusting more surfaces, she has 15 fingerprints in all. How many new fingerprints did she find?
2. **Model** $x + 8 = 15$
3. **Solve** $x = 15 + 8 = 23$
4. **Answer** 23 new fingerprints.

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** x is increased by 8. What operation undoes adding 8 — adding 8 again, or subtracting 8?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do you decide whether a word problem needs an equation or an inequality?

SENTENCE STARTER I use an equation when the problem says ___, but I use an inequality when the problem says ___ because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 8.7

Equations and Inequalities Problem Solving

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Write a model, solve, and check reasonableness for each problem.

Word Problem	Model	Solution	Reasonable?
Five identical boxes weigh a total of 60 pounds. How much does each box weigh?			
A shelf can hold at most 30 books. There are already 17 books. How many more can be added?			
A baker split 84 cookies equally into bags of 12. How many bags did she use?			

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each situation to its correct model

_____ Item 1 (A)

_____ Item 2 (B)

_____ Item 3 (C)

_____ Item 4 (D)

3 **Decide whether each pair balances. Check Yes or No, then explain.**

Solve this two-step problem: 3x + 5 = 26

Left side	Right side	Balanced?

EXPLAIN YOUR CHOICE

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A student solved 4n = 52 and got n = 13. Is the answer reasonable if n represents the number of notebooks in a box?

(A) No — 13 is too many notebooks

(B) No — the answer should be 48

(C) Yes — 13 notebooks per box is reasonable

(D) Yes — but only if n is a fraction

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Problem** Seven evidence lockers hold 84 case files in all, split equally. How many files per locker?
2. **Model** 7f = 84
3. **Solve** f = 84 − 7 = 77
4. **Answer** 77 files per locker.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do you decide whether a word problem needs an equation or an inequality?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 8.7

Equations and Inequalities Problem Solving

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Check each student's model. If wrong, write the correct model and solve.

Problem	Student's Model	Correct?	Fix (if needed)
A tank holds at most 500 gallons. It has 320. How many more gallons can be added?	320 + g = 500		
Six friends split a pizza bill equally. Each paid $8. What was the total?	t / 6 = 8		
A runner must complete more than 5 laps. She has done 2. How many more does she need?	2 + m = 5		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 **Find the mistake. Explain it, then show the correct work.**

Find the Problem-Solving Error

1. **Problem** A parking lot holds at most 200 cars. There are 134 already. How many more can park?
2. **Model** $134 + c = 200$
3. **Solve** $c = 66$
4. **Answer** Exactly 66 more cars can park.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 **Answer in complete sentences.**

Create two word problems about the same situation: one that requires an equation and one that requires an inequality. Solve both and explain why the models are different.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How do you decide whether a word problem needs an equation or an inequality?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help